A Data-Driven Study of Pollution Trends, Monitoring Stations, and Weather Associations using Python (2024-2025)

Delhi Air Quality Analysis | Python, Pandas, NumPy, Matplotlib, Seaborn | Google Colab

Python | Pandas | NumPy | Matplotlib | Seaborn | Google Colab

1. Project Overview

This project analyzes hourly air-quality and weather data from 39 monitoring stations across Delhi (2024-2025) using Python. The analysis covers data cleaning, exploratory data analysis (EDA), statistical analysis and visualization to understand how pollution changes by hour, day, month, season, station and weather condition.

The project aims to transform raw sensor readings into useful information that supports air-quality monitoring and better decision-making.

2. Data Source

Item	Details
Dataset Name	Combined Delhi Air Quality Dataset (39 stations, hourly)
Source	opencity - Open Data Portals (file hosted in the project's GitHub repository)
File Type	CSV
Location / Timeline	Delhi, India / 2024-2025
Domain	Environmental Data Analytics
Number of Records	683,803
Number of Attributes	30

The dataset contains hourly readings of pollutants (particulate matter, gases and volatile organic compounds) and weather variables (temperature, humidity, wind, rainfall, solar radiation and pressure) recorded at each station.

3. Problem Statement

Delhi's air quality is monitored continuously, but the raw data has many stations, many pollutants and many gaps, so it is hard to see the patterns that matter. Without proper analysis it is difficult to tell when, where and under what weather conditions pollution becomes dangerous.

This project analyzes the dataset to identify the key factors (time, season, location and weather) that influence air pollution in Delhi and to compare the observed levels with health guidelines.

4. Objectives

Data objectives

- To understand the dataset structure
- To clean and prepare the data for analysis
- To handle missing values and duplicates
- To perform Exploratory Data Analysis
- To create meaningful visualizations
- To generate actionable insights

Analytical objectives

- Analyze hourly, daily, monthly and yearly pollution trends
- Compare PM2.5 and PM10 across monitoring stations
- Examine pollution distributions, variability and outliers
- Investigate correlations between different pollutants
- Study relationships between pollution and weather variables
- Compare weekday and weekend pollution levels
- Develop data-driven insights to support air-quality monitoring

5. Attribute (Column / Feature) Details

Original columns (30)

Attribute Name	Type	Description
Station ID	Text	Unique identifier of the monitoring station (39 stations)
State	Text	State of the station (Delhi)
City	Text	City of the station (Delhi)
Station Name	Text	Name of the monitoring station
Timestamp	Datetime	Date and hour of the record
readings_in_hour	Integer	Number of readings combined into the hourly record (typically 4)
PM2.5 (µg/m³)	Float	Fine particulate matter (diameter up to 2.5 µm)
PM10 (µg/m³)	Float	Coarse particulate matter (diameter up to 10 µm)
NO (µg/m³)	Float	Nitric oxide
NO2 (µg/m³)	Float	Nitrogen dioxide
NOx (ppb)	Float	Total oxides of nitrogen
NH3 (µg/m³)	Float	Ammonia
SO2 (µg/m³)	Float	Sulphur dioxide
CO (mg/m³)	Float	Carbon monoxide
Ozone (µg/m³)	Float	Ground-level ozone
Benzene (µg/m³)	Float	Benzene (volatile organic compound)
Toluene (µg/m³)	Float	Toluene (volatile organic compound)
Xylene (µg/m³)	Float	Xylene (volatile organic compound, about 90% missing)
O Xylene (µg/m³)	Float	O-Xylene (completely empty, dropped)
Eth-Benzene (µg/m³)	Float	Ethylbenzene (about 85% missing)
MP-Xylene (µg/m³)	Float	M,P-Xylene (about 85% missing)
AT (°C)	Float	Air temperature
RH (%)	Float	Relative humidity
WS (m/s)	Float	Wind speed
WD (deg)	Float	Wind direction in degrees (0-360)
RF (mm)	Float	Rainfall
TOT-RF (mm)	Float	Total rainfall
SR (W/mt2)	Float	Solar radiation
BP (mmHg)	Float	Barometric (atmospheric) pressure
VWS (m/s)	Float	Vertical wind speed (about 86% missing)

Engineered features (created in Section 7.4)

Feature	Description
hour, day, month, year	Time parts extracted from Timestamp
day_of_week	Day of the week (0 = Monday ... 6 = Sunday)
is_weekend	1 for Saturday and Sunday, otherwise 0
Season	Winter (Dec-Feb), Summer (Mar-May), Monsoon (Jun-Sep), Post-Monsoon (Oct-Nov)

6. Tools & Technologies

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

7. Data Pre-Processing and Feature Engineering

The dataset was loaded, inspected and then prepared for analysis. The initial inspection (7.2) is done first because the cleaning decisions in 7.3 depend on it.

7.1 Importing Libraries and Loading the Data

7.2 Initial Data Inspection

7.3 Data Cleaning

Data cleaning steps

- Checked data types of all columns
- Identified missing values
- Handled null values
- Removed duplicate records (none found)
- Corrected inconsistent data (blank strings converted to missing values)
- Converted data types where required (Timestamp to datetime)

Step 1: Data types

Step 2: Missing values

- Pollutant columns were filled with the median of the same station.
- Columns with less than 30% missing values were filled with the overall median.
- Air temperature and pressure were interpolated within each station (up to 8 hours), then filled with the station median.
- `0 Xylene` is 100% empty, so it was dropped. Xylene, Eth-Benzene, MP-Xylene and VWS (more than 85% missing) were left unfilled rather than inventing values.

Step 3: Blank strings

7.4 Feature Engineering

New features were created to improve the analysis:

- Time features (hour, day, day of week, month, year, weekend flag) from the Timestamp
- Season, based on the Delhi climate calendar
- Further analysis-specific features are created inside the charts: CPCB air-quality category from daily mean PM2.5, PM2.5/PM10 ratio per station, rain-level groups, 16 wind-direction sectors and weather bins

Wind direction, rainfall and weather bins are analyzed on the original (unfilled) data so that median imputation does not distort the results.

8. Analysis and Visualizations

The dataset was analyzed using Exploratory Data Analysis techniques to understand patterns and relationships. The analysis is organized into statistical analysis (Section 10) and 15 visualizations (Section 11): 3 univariate, 8 bivariate and 4 multivariate. Every chart is followed by a short interpretation.

9. Exploratory Data Analysis (EDA)

Initial inspection (code in Section 7.2) included:

- **Data frame structure:** 683,803 rows and 30 columns (24 float, 1 integer, 5 text columns; Timestamp was stored as text and converted to datetime)
- **Statistical summary:** `describe()` table of all numerical columns (minimum, maximum, quartiles)
- **Null values:** PM2.5 (about 6% missing) and PM10 (about 5%) are mostly complete, Toluene is about 33% missing, and Xylene, Eth-Benzene, MP-Xylene and VWS are more than 85% missing
- **Duplicate records:** 0 duplicate rows
- **Coverage:** 39 unique monitoring stations

EDA helped in understanding the distribution and characteristics of the data and guided the cleaning strategy in Section 7.3.

10. Statistical Analysis

Statistical measures were used to summarize the dataset: mean, median and mode (below), minimum and maximum values (`describe()` table in Section 7.2), variance, standard deviation, skewness and kurtosis.

Key statistics (cleaned data)

Variable	Mean	Median	Std. dev.	Skewness
PM2.5 (µg/m³)	99.48	68.77	93.73	2.36
PM10 (µg/m³)	204.04	171.00	140.63	1.50
NO2 (µg/m³)	44.38	36.47	34.93	2.47
CO (mg/m³)	1.27	1.03	1.02	5.29
Ozone (µg/m³)	34.06	20.42	37.75	2.38
AT (°C)	26.35	27.70	6.51	-0.79
RH (%)	64.06	65.25	17.80	-0.40
WS (m/s)	0.97	0.77	0.90	7.15

Interpretation:

PM2.5 has a mean of 99.5 µg/m³ but a median of only 68.8, and PM10 has a mean of 204.0 against a median of 171.0. The mean sits above the median because of extreme winter hours.

Both are right-skewed (skewness 2.36 for PM2.5 and 1.50 for PM10) with wide variability (standard deviation 93.7 and 140.6 µg/m³).

Temperature averages 26.3 °C (standard deviation 6.5) and humidity 64%. The mode is less meaningful for continuous sensor readings, so the median is used as the typical value.

11. Data Visualization

Visualization techniques were used to represent the data graphically and make patterns easier to understand.

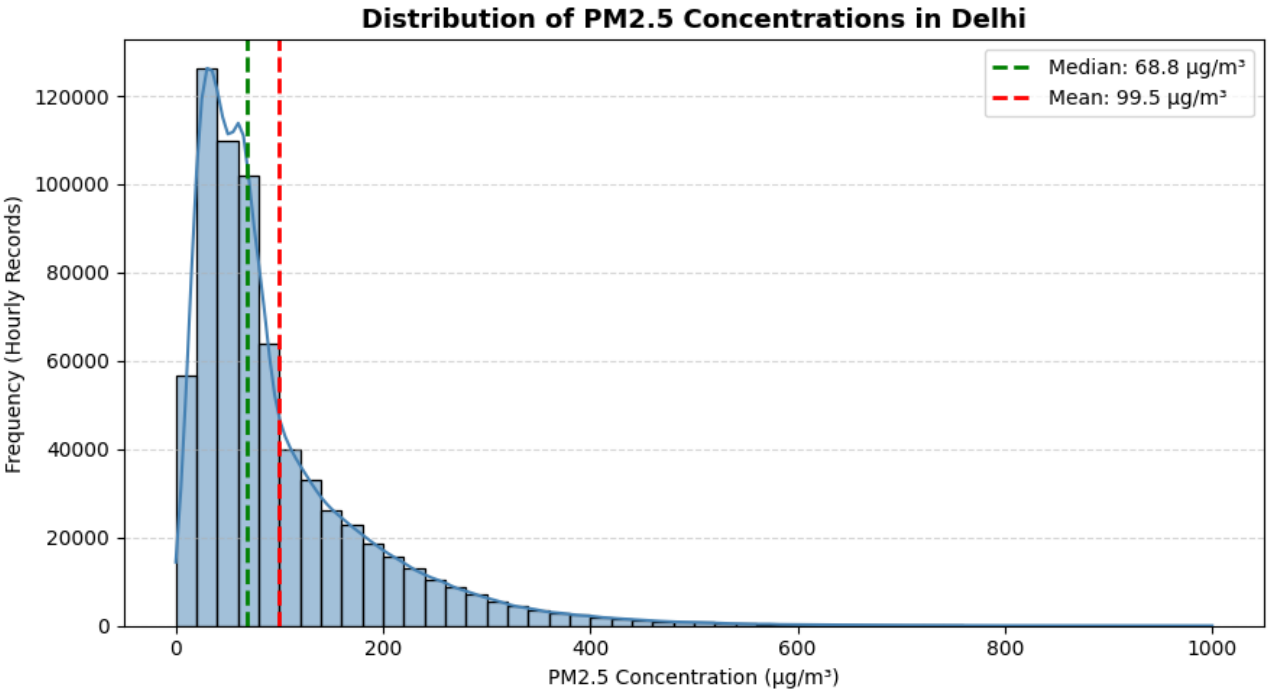
11.1 Univariate Analysis

Univariate analysis was performed to analyze one variable at a time.

Analysis performed:

- Distribution of PM2.5 (Chart 1)
- Outliers and variability of PM2.5, PM10 and NO (Chart 2)
- Average concentration of each pollutant against WHO guidelines (Chart 3)

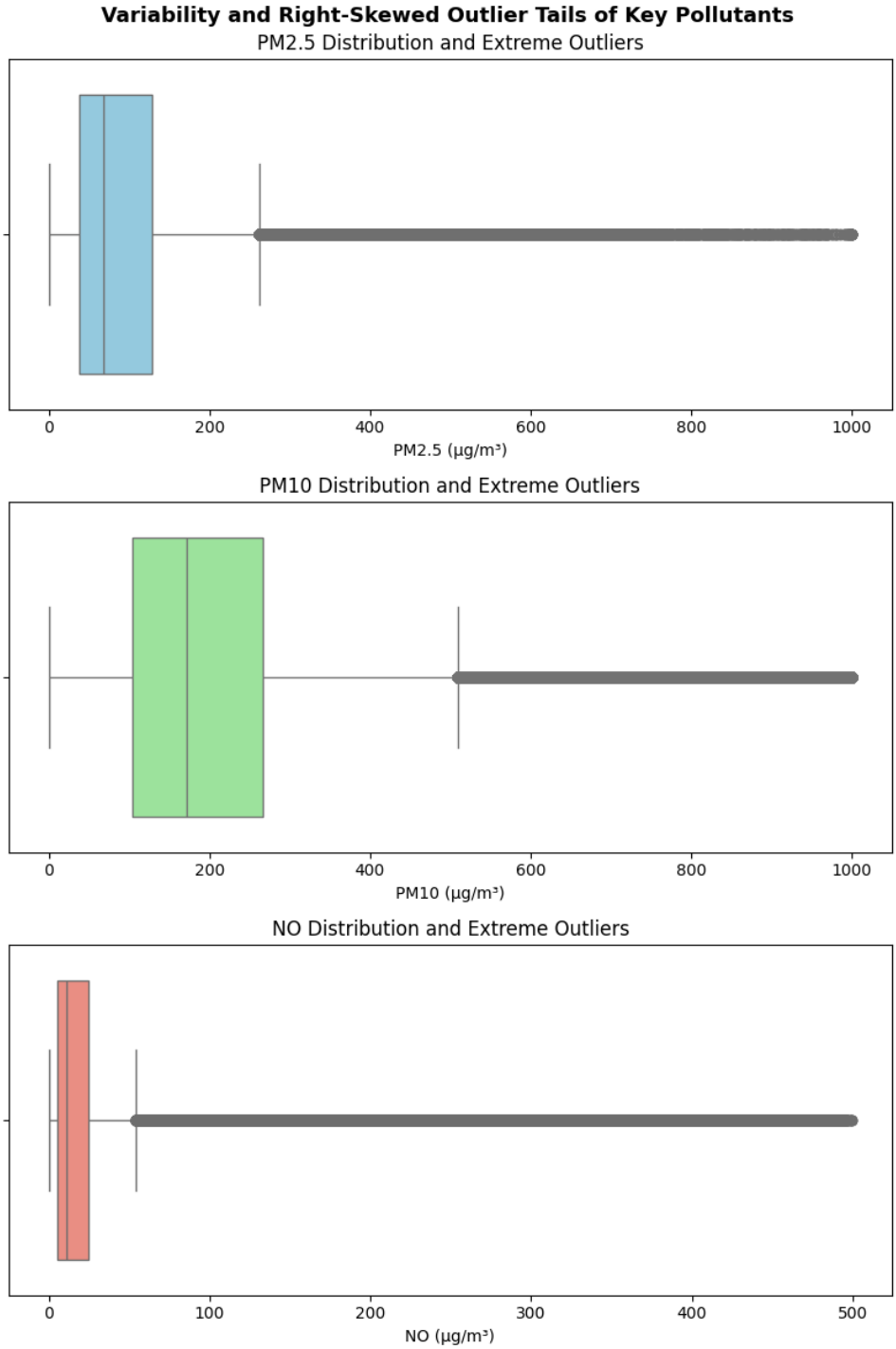
Chart 1: Distribution of PM2.5



Interpretation: PM2.5 is right-skewed: most hours are low (peak around 25 to 50 µg/m³), with a long tail of extreme hours reaching 600 µg/m³ and beyond.

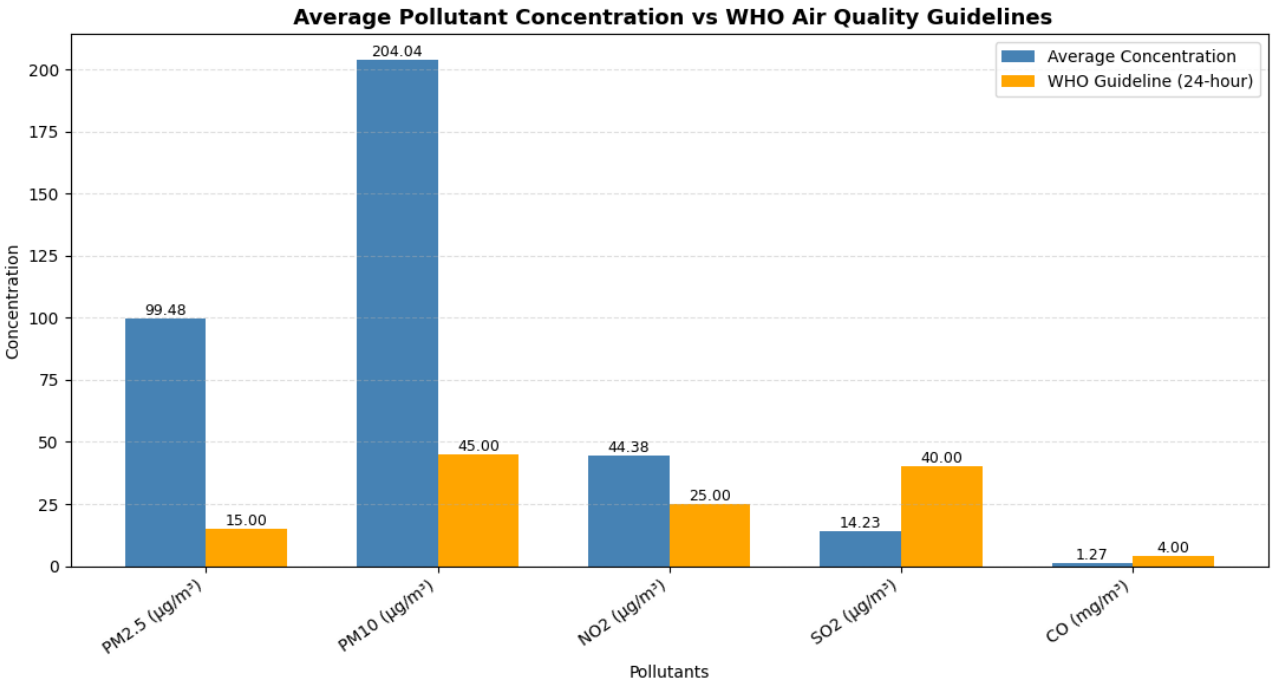
The mean (99.5 µg/m³) is well above the median (68.8 µg/m³), so the median is the better "typical" value and the mean shows overall exposure. Together with the monthly category chart (Chart 7), the tail points to winter Very Poor and Severe episodes.

Chart 2: Outliers and variability of key pollutants



Interpretation: All three pollutants are strongly right-skewed. Median PM2.5 is about **68.77 µg/m³** (IQR about 38 to 128) and median PM10 is about 172 µg/m³ (IQR about 105 to 267), while NO is much lower (median about 12 µg/m³, IQR about 5 to 25). A dense run of extreme readings extends to 1000 µg/m³ for PM and 500 µg/m³ for NO, where values stack up at the instrument ceiling. These extremes correspond to winter pollution episodes, so they are retained and the median is used as the typical value.

Chart 3: Average pollutant concentration vs WHO guidelines



Interpretation:

- **PM2.5 & PM10:** Average PM2.5 (99.48 µg/m³) and PM10 (204.04 µg/m³) significantly exceed WHO air quality guidelines (15 µg/m³ and 45 µg/m³ respectively). This corresponds to concentrations that are **6.6×** and **4.5×** higher than safe limits.
- **NO2:** The average concentration (44.38 µg/m³) also exceeds the WHO recommended limit of 25 µg/m³.
- **SO2 & CO:** Both SO2 (14.23 µg/m³) and CO (1.27 mg/m³) remain safely within their respective WHO guidelines (40 µg/m³ and 4 mg/m³).
- **Conclusion:** Fine and coarse particulate matter (PM2.5 and PM10) represent the most severe and chronic air pollution threats in Delhi.

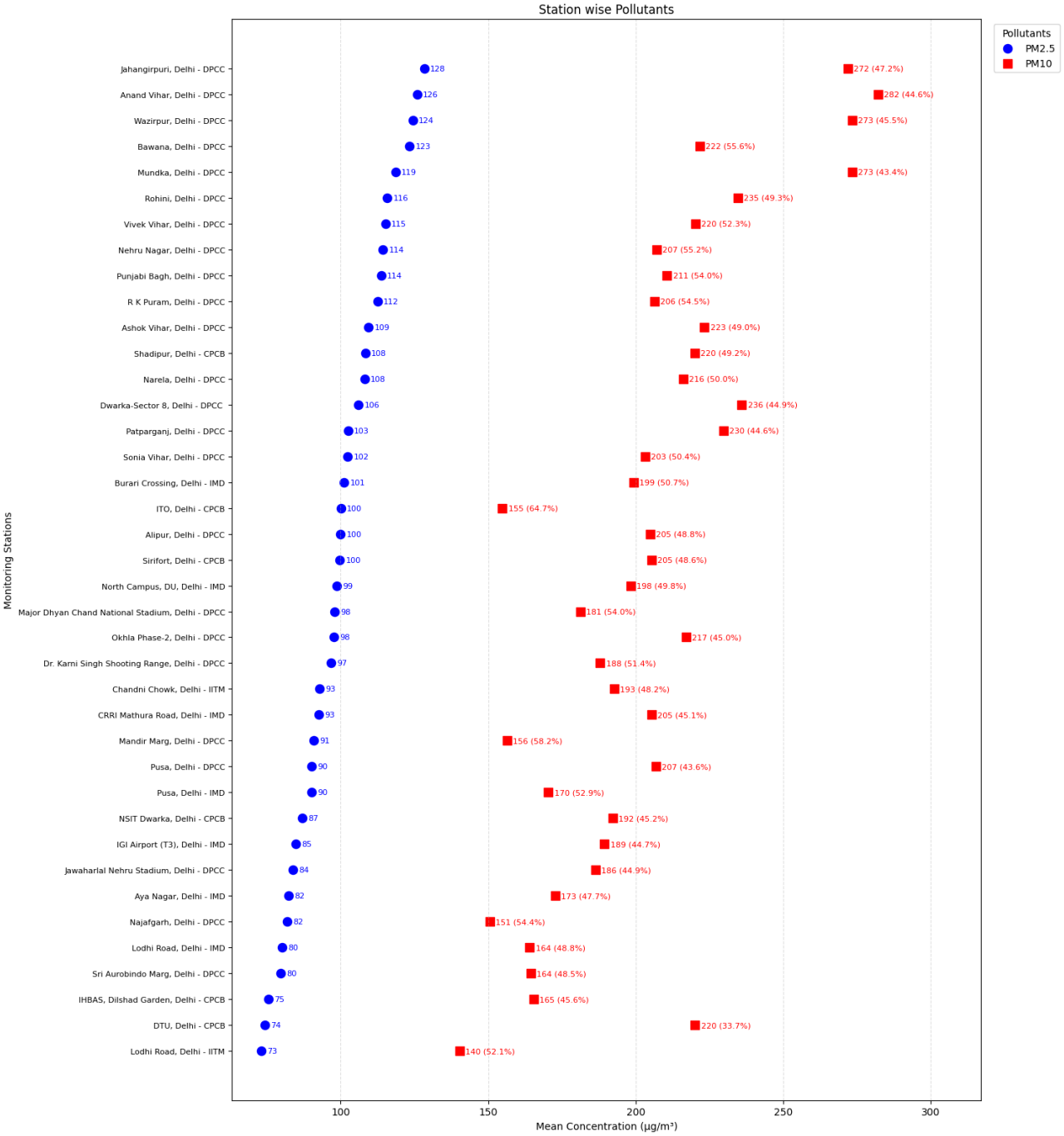
11.2 Bivariate Analysis

Bivariate analysis was performed to study relationships between two variables.

Analysis performed:

- Station vs PM2.5 and PM10 (Chart 4)
- Month vs PM2.5 and PM10 by year (Chart 5)
- Date vs daily PM2.5 and PM10 (Chart 6)
- Month vs air-quality category (Chart 7)
- Season vs pollutants (Chart 8)
- Day of week vs PM2.5 (Chart 9)
- Wind direction vs PM2.5 (Chart 10)
- Wind speed, humidity, solar radiation and pressure vs PM2.5 (Chart 11)

Chart 4: Station-wise PM2.5 and PM10



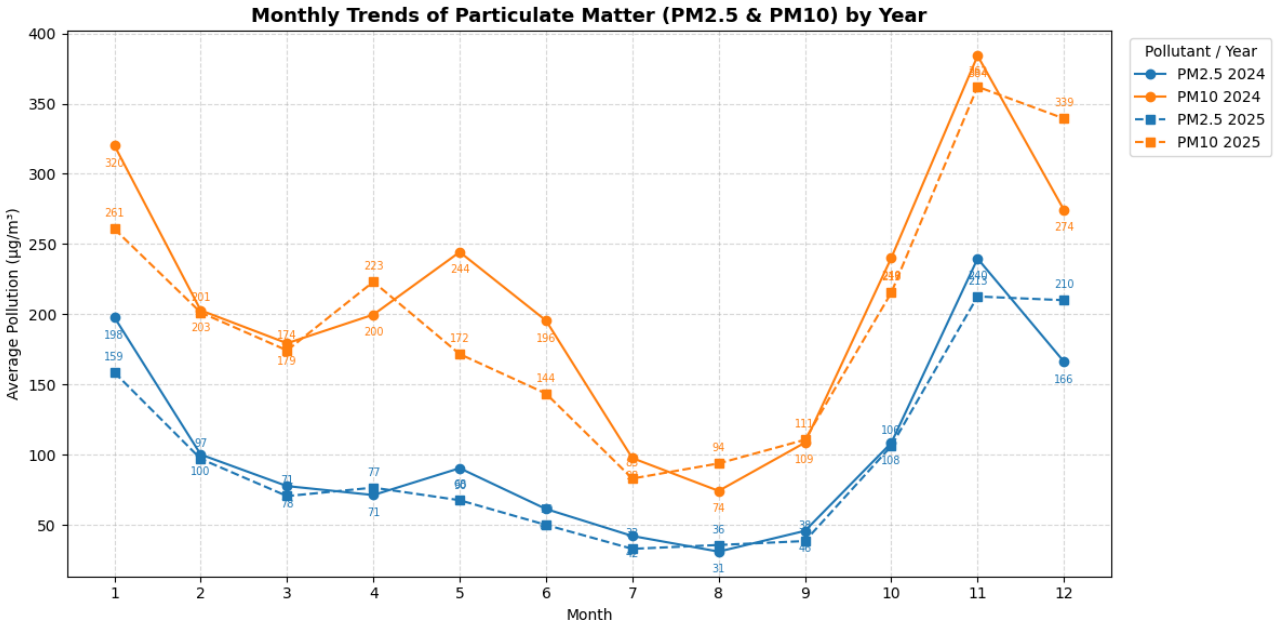
Interpretation:

For PM2.5, Jahangirpuri (~128.4 µg/m³), Anand Vihar (~125.9) and Wazirpur (~124.5) are the most polluted stations.

For PM10, Mundka is the highest (~273.5 µg/m³) and Lodhi Road (IITM) is the lowest (~140.2).

ITO has the highest PM2.5/PM10 ratio (64.7%; PM2.5 about 100, PM10 about 155), which suggests fine particles from traffic and combustion. DTU has the lowest (33.7%; PM2.5 about 74, PM10 about 220), which suggests coarse dust.

Chart 5: Monthly trends of particulate matter by year

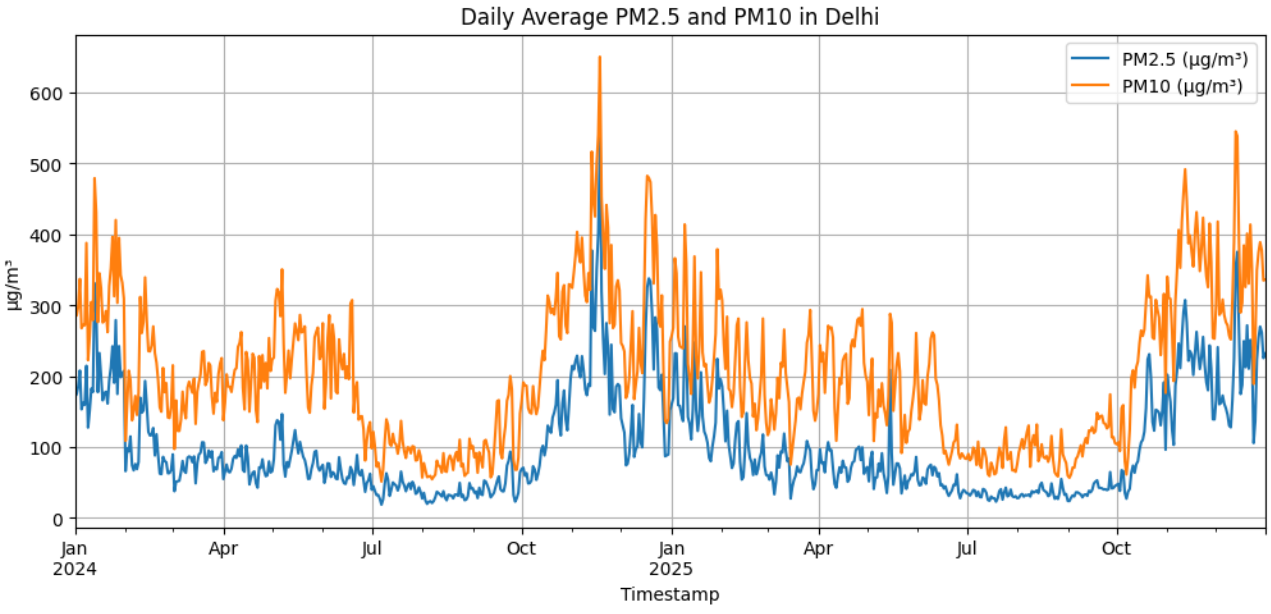


Interpretation:

Particulate concentrations are lowest during the monsoon and late summer, bottoming out in August (Month 8) with average PM2.5 at **33.30 µg/m³** and PM10 at **83.97 µg/m³**.

- **Winter Peak:** Pollution climbs sharply from October (PM2.5 about 107) and peaks in November (Month 11) with average PM2.5 of **226.15 µg/m³** and PM10 of **373.09 µg/m³**. December and January stay high (PM2.5 about 188 and 178).
- **Magnitude:** The November peak is **6.8x (PM2.5)** and **4.4x (PM10)** higher than the August low.
- **Particle mix:** PM10 is about 1.6 to 3.0 times PM2.5 across the year. The ratio is lowest in winter (about 1.6x, mostly fine particles from combustion) and highest in April to June (about 2.6 to 3.0x), pointing to more coarse dust in the hot, dry months.

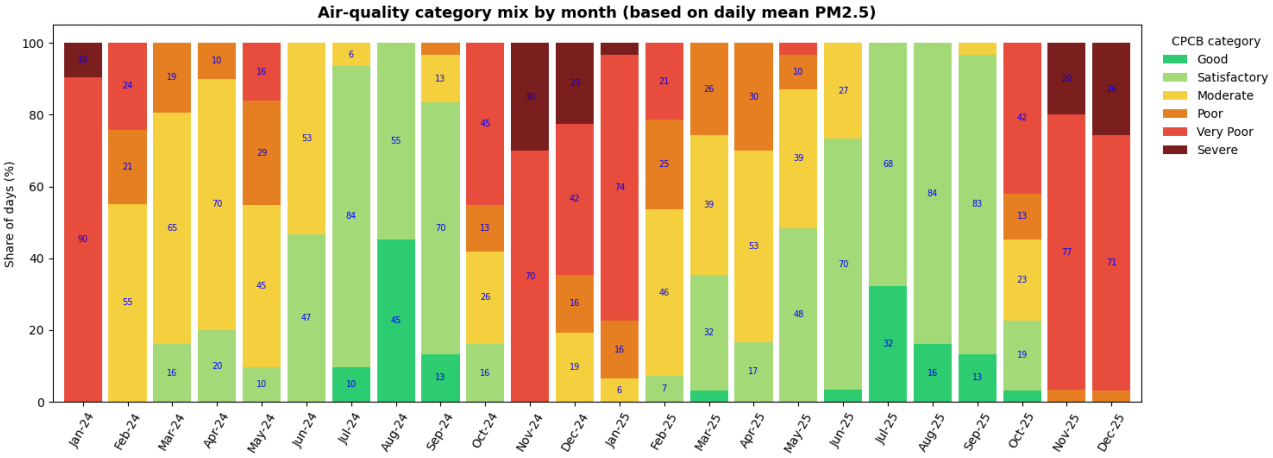
Chart 6: Daily trend of PM2.5 and PM10



Interpretation:

- **Daily Ranges:** Daily aggregated PM2.5 averages range from a clean baseline of approximately **17.9 µg/m³** (recorded on July 7, 2024) to winter peaks exceeding **310.0 µg/m³**.
- **PM10 Variations:** Daily aggregated PM10 averages range from a minimum of approximately **44.9 µg/m³** during peak monsoon up to extreme winter peaks of **650.0 µg/m³**.
- **Extreme Events:** The lowest single-day clean period occurred on July 7, 2024 (PM2.5: ~17.9 µg/m³), whereas the highest daily average pollution episodes are clustered around late November and mid-December.

Chart 7: Air-quality category mix by month



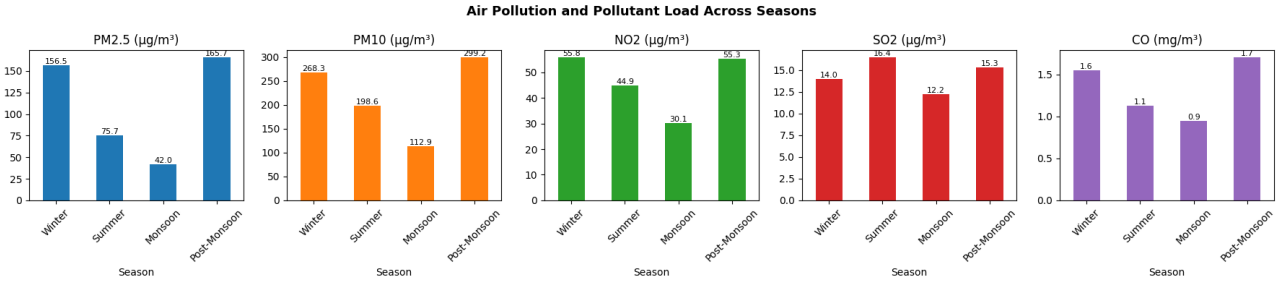
Interpretation:

Winter is the worst season in both years. Nov-24 had 70% Very Poor + 30% Severe days, and Nov-25 had 77% Very Poor + 20% Severe, so nearly every winter day was Very Poor or worse.

Monsoon (Jul to Sep) is the cleanest. Aug-24 had 45% Good + 55% Satisfactory, and Jul-25 had 32% Good + 68% Satisfactory, with no day worse than Satisfactory.

Spring and early summer sit in the middle. Mar to May are mostly Moderate or Poor (Mar-24 65% Moderate, Apr-25 53% Moderate + 30% Poor), so winter pollution is what pushes the yearly averages above the limits.

Chart 8: Seasonal variation of pollutants

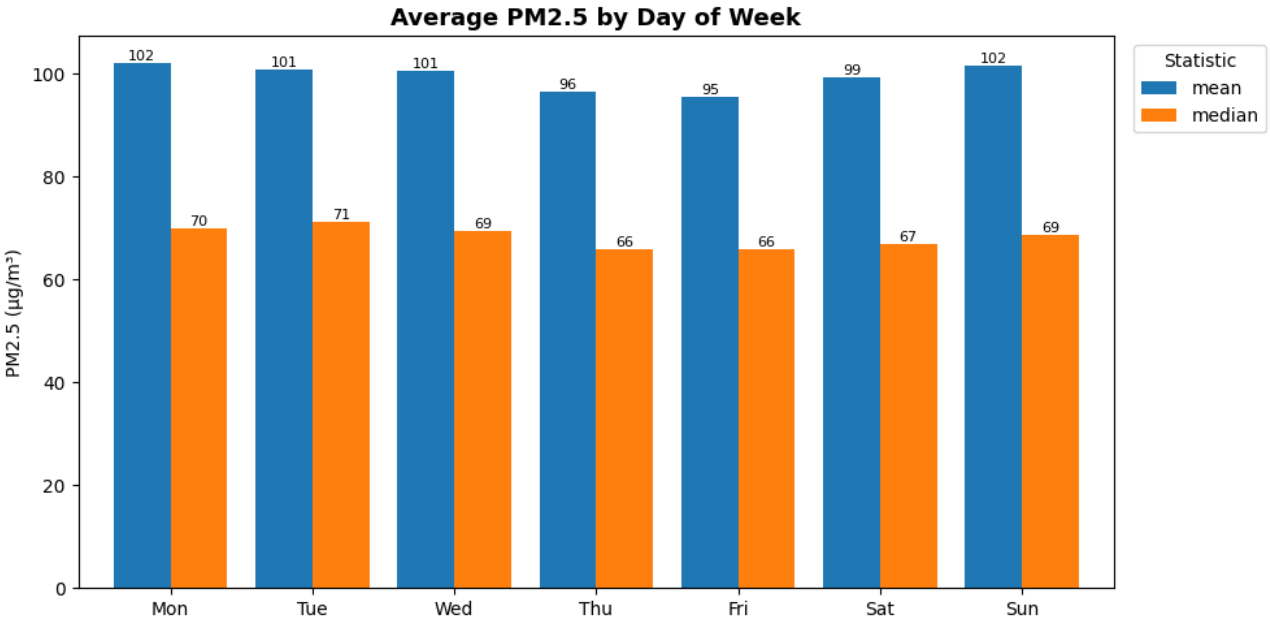


Interpretation:

- **Winter & Post-Monsoon Peaks:** Post-Monsoon and Winter exhibit extreme levels of particulate matter. Average PM2.5 peaks at **165.7 µg/m³** in Post-Monsoon and **156.5 µg/m³** in Winter. PM10 follows the same trend, peaking at **299.2 µg/m³** in Post-Monsoon and **268.3 µg/m³** in Winter.

- **Monsoon Scavenging:** Monsoon is the cleanest season by far, with PM2.5 dropping to **42.0 µg/m³** and PM10 to **112.9 µg/m³**, showcasing the highly effective atmospheric wet-deposition cleansing effect.
- **Gas Pollutant Patterns:** NO2 and CO are highly elevated in Winter and Post-Monsoon due to stagnant air, whereas SO2 exhibits a unique peak in Summer (**16.4 µg/m³**), likely tied to higher power-generation and industrial coal-burning demands.

Chart 9: Weekday vs weekend (PM2.5 by day of week)



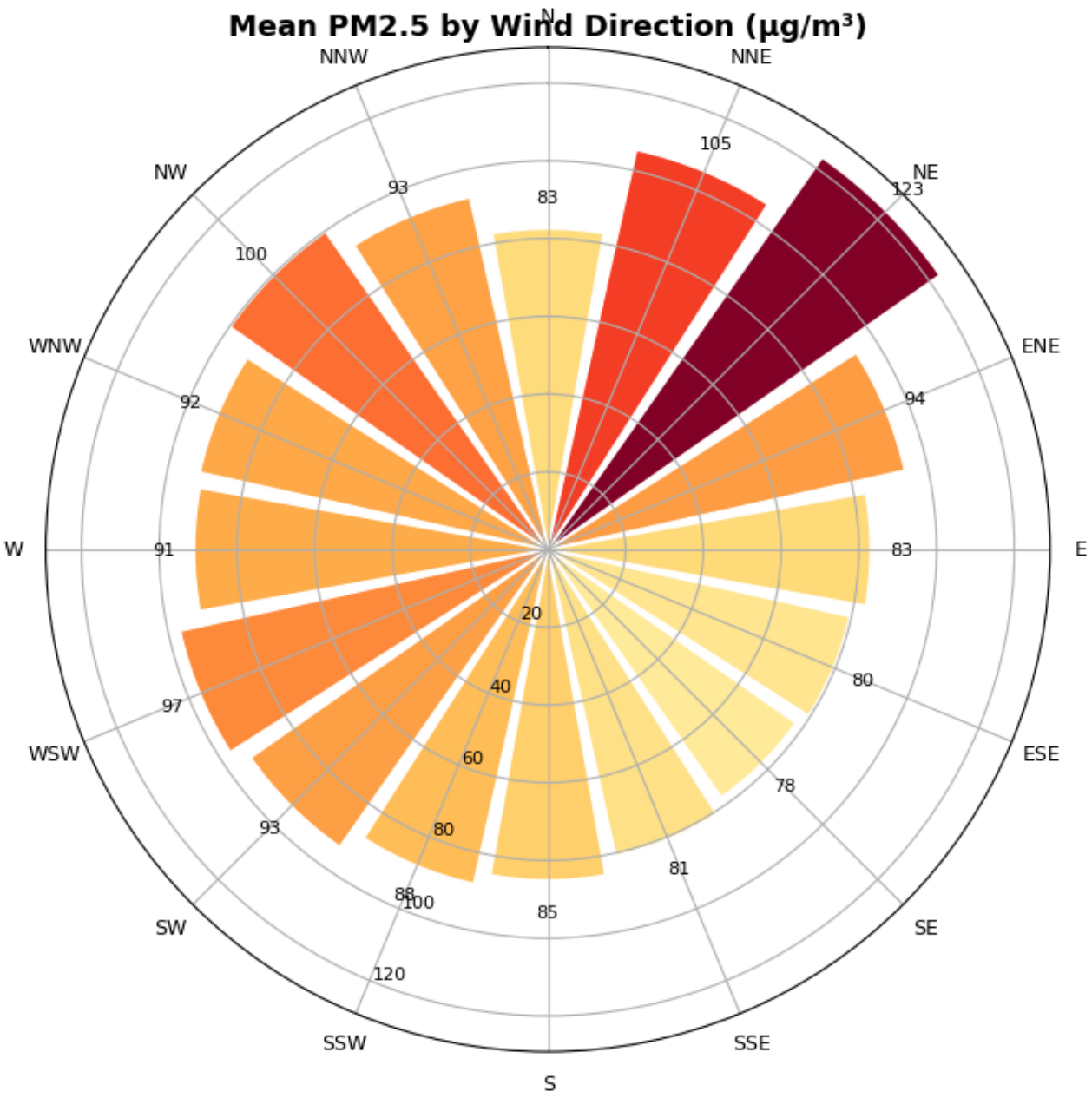
Interpretation: PM2.5 is almost flat across the week. Daily means range only from 95 µg/m³ (Fri) to 102 (Mon and Sun), a spread of about 7 µg/m³ (roughly 7%).

Medians tell the same story. They sit between 66 (Thu and Fri) and 71 (Tue), and the mean is above the median every day (for example Mon 102 vs 70) because of extreme winter hours.

No weekend effect. Sat (99, median 67) and Sun (102, median 69) are in line with weekdays (Mon to Fri mean about 99), so there is no sign that lower weekend traffic cleans the air.

Pollution is regional and weather-driven, not tied to the working week. Thu and Fri are the slightly cleanest days, but the gap is too small to act on, and every day stays above the 40 µg/m³ CPCB annual limit.

Chart 10: PM2.5 by wind direction

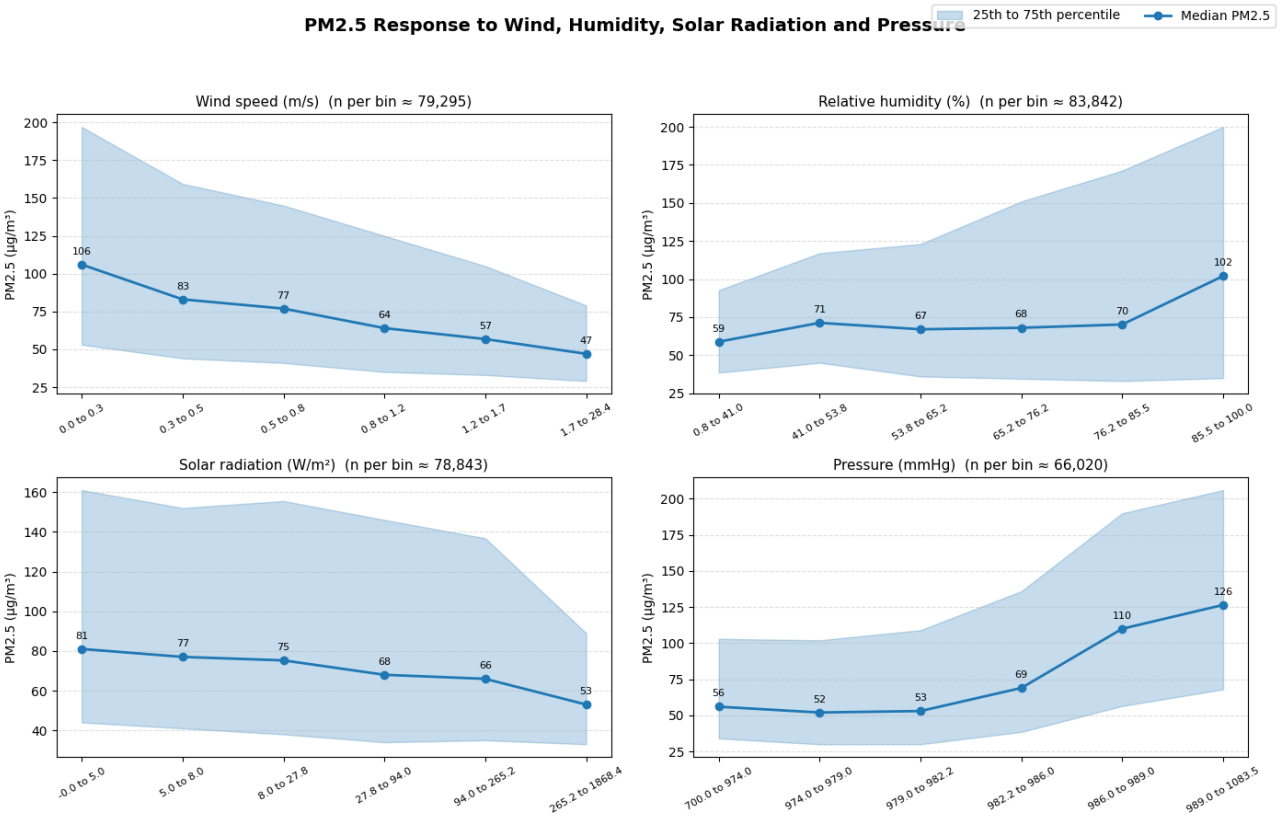


Interpretation: Winds from the north-east carry the most polluted air (about 123 µg/m³ PM2.5), followed by NNE (105) and NW (100).

The south-east sectors bring the cleanest air (about 78 µg/m³), so the worst direction is about 1.6× the cleanest.

The west and south-west sectors (WSW 97, SW 93) sit in the middle.

Chart 11: PM2.5 across weather bins



Interpretation: PM2.5 decreases from 106 to 47 µg/m³ as wind speed increases, indicating better pollutant dispersion.

Higher humidity is associated with increased PM2.5, reaching 102 µg/m³.

Increasing solar radiation shows a gradual decrease in PM2.5 from 81 to 53 µg/m³.

Higher atmospheric pressure is associated with increased PM2.5, reaching 126 µg/m³, suggesting a strong relationship with pollution levels.

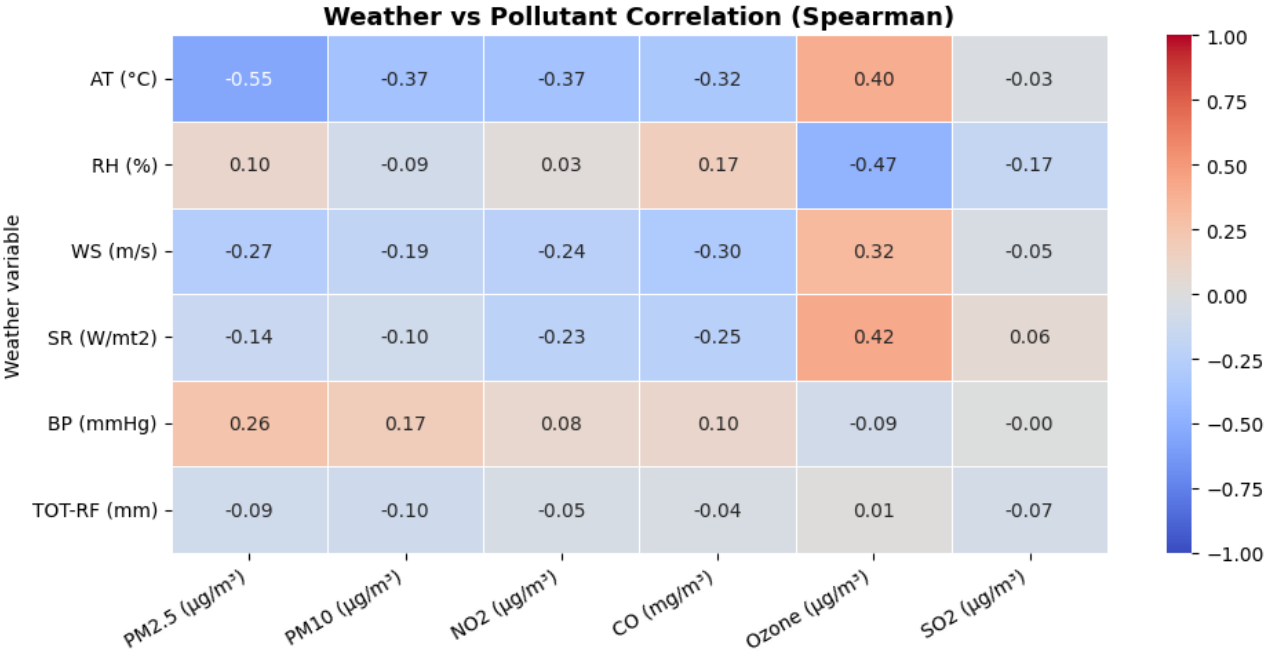
11.3 Multivariate Analysis

Multivariate analysis was used to analyze relationships among multiple variables.

Analysis performed:

- Correlation analysis: weather variables vs pollutants (Chart 12)
- Hour of day, month and PM2.5 together (Chart 13)
- Rain level, station, month and particulate matter together (Chart 14)
- Hour of day and six gaseous pollutants together (Chart 15)

Chart 12: Correlation between weather variables and pollutants

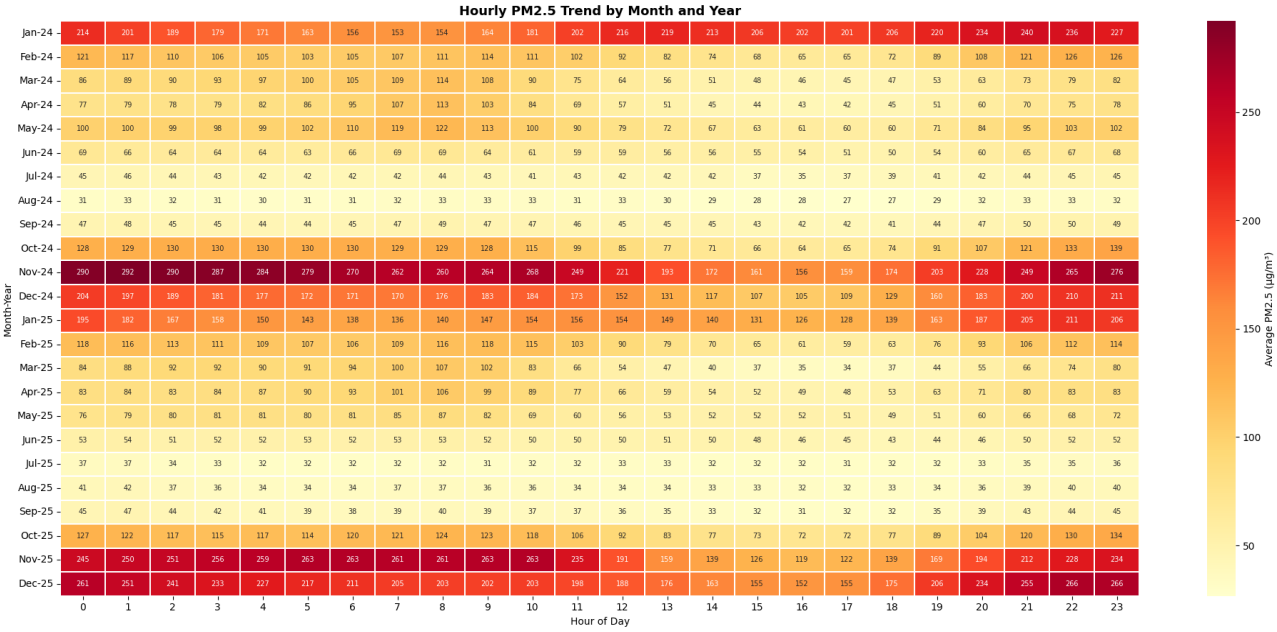


Interpretation: Temperature has a negative link with particulate pollution. AT vs PM2.5 is -0.44 and AT vs PM10 is -0.31, so colder days have higher PM2.5 and PM10. This matches the winter peaks in the monthly trend chart (Chart 5).

PM2.5 and PM10 are strongly correlated (0.82). They rise and fall together, which suggests common sources and the same weather drivers.

CO and NO2 show moderate positive links with particulates. CO is 0.45 with PM2.5 and 0.46 with PM10, and NO2 is 0.33 and 0.38. This points to combustion and traffic emissions as a shared source.

Chart 13: Hourly PM2.5 trend by month and year

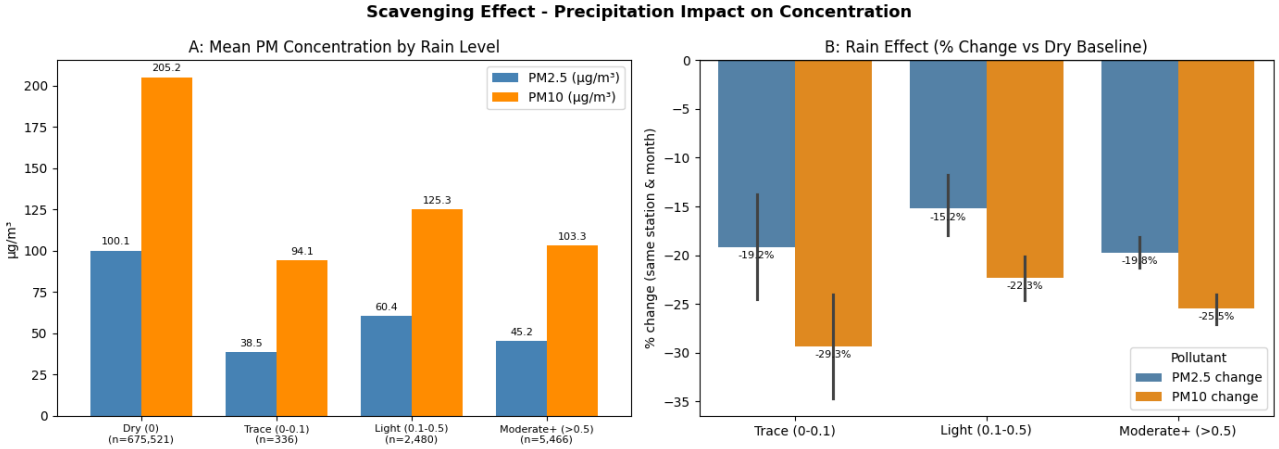


Interpretation:

PM2.5 follows a daily cycle: high at night and early morning, lowest in the afternoon (14 to 17h), rising again from 19h. In Nov-24 it falls from 292 µg/m³ (1h) to 156 (16h), a drop of about 47%, while monsoon months stay flat (Aug-24: 27 to 33).

The peak-to-lowest gap is about 10.8x (Nov-24 1h vs Aug-24 17h). Winter (Oct to Jan) is the critical period, and Nov and Dec carry the highest night and evening exposure, so interventions and alerts should target 19h to 9h. 2025 is lower than 2024 in most months, but Dec-25 is worse than Dec-24 (261 vs 204 at 0h).

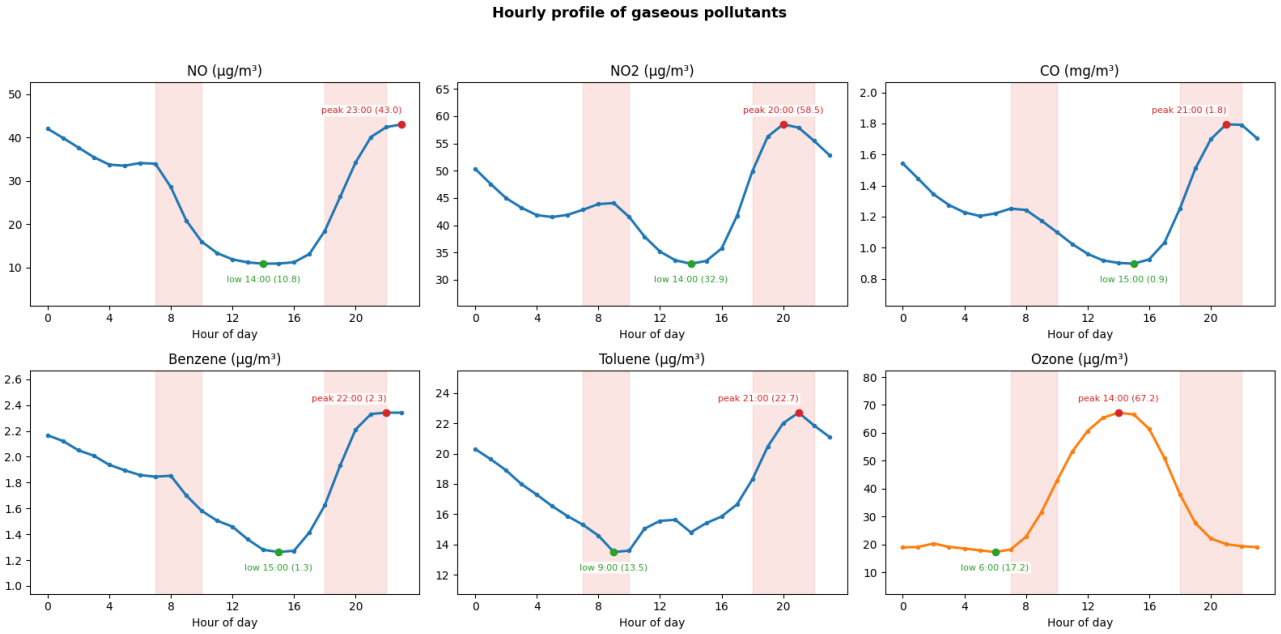
Chart 14: Rain effect on particulate matter



Interpretation:

- **Washout Efficiency (Plot A):** Hourly periods with precipitation are highly scarce but show rapid, significant reductions. Dry hours average **100.1 µg/m³** for PM2.5 and **205.2 µg/m³** for PM10. During active moderate+ rain (>0.5 mm), average PM2.5 drops to **45.2 µg/m³**.
- **Relative Impact vs Dry Baselines (Plot B):** Standardizing each wet event against its same-station, same-month dry baseline confirms immediate relative drops:
- **Trace rain (0-0.1 mm)** causes a relative washout of **-19.2% for PM2.5** and **-29.3% for PM10**.
- **Light rain (0.1-0.5 mm)** results in a relative drop of **-15.2% for PM2.5** and **-22.3% for PM10**.
- **Moderate+ rain (>0.5 mm)** maintains a relative reduction of **-19.8% for PM2.5** and **-25.5% for PM10**.
- **Insight:** Coarse particles (PM10) are consistently washed out more effectively than fine aerosol particles (PM2.5) across all precipitation intensities due to their larger cross-sectional area and higher collision efficiency with falling raindrops.

Chart 15: Daily rhythm of gaseous pollutants



Interpretation:

The traffic gases peak in the evening and night, not in the morning. NO peaks at 23:00 (43.0 µg/m³), NO2 at 20:00 (58.5), CO at 21:00 (1.8 mg/m³), Benzene at 22:00 (2.3) and Toluene at 21:00 (22.7). The morning rush (7 to 10h) shows only a weak bump (NO2 about 44 at 8 to 9h, CO about 1.25 at 7h).

All of them drop to a low in the afternoon.

12. Summary of Findings

- **Particulate matter far exceeds safe limits:** average PM2.5 is 99.48 µg/m³ and PM10 is 204.04 µg/m³, about 6.6 and 4.5 times the WHO 24-hour guidelines (15 and 45 µg/m³). NO2 (44.38 µg/m³) also exceeds its limit, while SO2 and CO stay within theirs.
- **Winter is the worst season:** November peaks at 226.15 µg/m³ PM2.5, 6.8 times the August low (33.30). In Nov-24 and Nov-25 nearly every day was Very Poor or Severe, while the monsoon months were the cleanest.
- **Hotspot stations:** Jahangirpuri, Anand Vihar and Wazirpur have the highest PM2.5 (about 125 to 128 µg/m³) and Mundka has the highest PM10 (about 274 µg/m³).
- **Time of day matters:** PM2.5 is highest at night and early morning and lowest in the afternoon, and the traffic-related gases peak in the evening and night (NO2 at 20:00, CO at 21:00).
- **Weather drives pollution:** colder, calmer air traps pollutants (AT vs PM2.5 correlation -0.44; PM2.5 falls from 106 to 47 µg/m³ as wind speed rises), and rain lowers PM2.5 by about 15 to 20% and PM10 by about 22 to 29%.
- **Wind direction:** north-east winds carry the most polluted air (about 123 µg/m³ PM2.5), about 1.6 times the cleanest direction.
- **No weekday-weekend effect:** daily mean PM2.5 stays between 95 and 102 µg/m³ across the week, so pollution is regional and weather-driven, not tied to the working week.

13. Types of Analysis

1. Descriptive Analysis

Descriptive analysis summarizes historical data.

- Average PM2.5 (99.48 µg/m³) and PM10 (204.04 µg/m³) compared with WHO guidelines
- Monthly, seasonal and station-wise pollution levels, with November peaking at 226.15 µg/m³ PM2.5
- Distribution and outliers of PM2.5, PM10 and NO

2. Diagnostic Analysis

Diagnostic analysis explains why something happened.

- Low wind speed, low temperature and stagnant air trap pollutants, which explains the winter peaks
- Rain washes particles out of the air, with a stronger effect on coarse PM10
- Traffic-related gases peak in the evening and night rather than in the morning rush hour

3. Predictive Analysis

Predictive analysis helps forecast future outcomes.

- The strong, repeating seasonal and hourly patterns give a baseline for anticipating severe winter spikes in the late-year months
- No forecasting model is built in this project; a PM2.5 forecast is listed under Future Enhancements

4. Prescriptive Analysis

Prescriptive analysis suggests actions based on insights.

- Target regulatory action at chronic hotspot stations such as Jahangirpuri and Anand Vihar
- Focus alerts and interventions on the high-risk window from 19:00 to 09:00 and on the winter months
- Standardize the monitoring pipeline to resolve data gaps in the trace gases

14. Future Enhancements and Suggestions

- Include more years of data and station coordinates for better accuracy and spatial (map) analysis
- Apply machine learning models (for example, time-series forecasting of PM2.5 or AQI category classification)
- Build interactive dashboards (Power BI, Tableau or Streamlit)
- Automate the reporting process so the analysis refreshes with new data
- Add external factors such as traffic, crop-residue burning and industrial emissions

15. Conclusion

The project successfully analyzed Delhi's hourly air-quality data using Python and extracted useful insights. Average PM2.5 (99.48 µg/m³) and PM10 (204.04 µg/m³) are far above WHO health guidelines, making particulate matter the region's most acute air-quality problem.

Data preprocessing and visualization helped in understanding patterns and relationships: pollution peaks in post-monsoon and winter (especially November), at night and early morning, and under cold, low-wind conditions, while monsoon rain cleans the air. It shows no weekday-weekend difference.

The results of this project can help authorities and monitoring agencies make better decisions by targeting chronic hotspot stations and high-risk hours, and by closing the data gaps in the trace-gas measurements.